TRUNG DANG

Curriculum Vitae

Department of Computer Science
Purdue University
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My Website
GitHub

Education

Aug 2019 – May 2023 **BSc, Computer Science**, *Purdue University*, West Lafayette, IN.
Algorithm Design, Analysis, and Implementation. Randomized Algorithm. Algorithmic Economics. Intro to Artificial Intelligence. Machine Learning.

Publications and Submissions

In Conference Proceedings

2021 Zitao Li, Trung Dang, Tianhao Wang, and Ninghui Li. MGD: A Utility Metric for Private Data Publication. In *8th International Conference on Networking, Systems and Security*, 8th NSysS 2021, page 106–119, New York, NY, USA, 2021. Association for Computing Machinery.

Under Submissions

2022 Trung Dang, Jasper C.H. Lee, Maoyuan Song, and Paul Valiant. Neighborhood optimality in mean estimation: beyond worst-case, beyond sub-Gaussian, and beyond $1 + \alpha$ moments. 2022. Under submission to *14th Innovations in Theoretical Computer Science Conference, ITCS 2023*.

Research Experience

Purdue University

Oct 2019 – Jan 2020 ***MGD: A Utility Metric for Private Data Publication***, *Published*, Advisor: Dr. Ninghui Li.
Inspired by the recent effort of synthesizing data with differential privacy guarantee from real-life data to protect users' private information, we introduced the MGD metric for calculating the utility of the synthetic dataset, which is based on the approximate version of the earth mover's distance. This new metric enjoys better range query error than other existing metrics. We also introduced a min-cost flow algorithm that computes this proposed metric in an efficient manner. This proposal helped us win the NIST "A Better Meter Stick for Differential Privacy" competition, and the paper was published as a proceeding to the NSysS 2021 conference.

Oct 2021 – present ***Probabilistic Click-Through Rate***, *In progress*, Advisor: Dr. Alex Psomas.
At a high level, we are given n boxes, each with a hidden reward from an unknown distribution, and an unbiased estimate for each reward, we want to choose k boxes that maximize the highest reward. This model reflects various real-life situations; for example, in the context of online ad auction, one can think of each reward as the unknown probability that a user clicks on an ad, and the unbiased estimate as a machine learning prediction. We have solved and analyzed the optimal strategy for picking the highest reward in various specific cases, and we are in the progress of proving reward guarantees of an algorithm class in the generic settings.

Jan 2022 – Aug 2022 ***Neighborhood Optimality in Mean Estimation***, *Submitted*, Advisor: Dr. Paul Valiant.
Mean estimation is among one of the most fundamental problems in statistics: given samples from an unknown distribution p and parameters δ and ϵ, we aim to estimate the mean of p such that the estimation is within ϵ distance away from the true mean with probability at least $1 - \delta$. Motivated by the recent effort of "beyond worst-case analysis" in the theoretical community, we introduced the notion of "neighborhood optimality" in the mean estimation problem, which smoothly extrapolates between the "instance optimality" and the "admissibility" notions. We then showed that the classic median-of-means algorithm is neighborhood optimal for a well-defined neighborhood structure, which implies that this algorithm guarantees good performance on all input (which contrasts against other algorithms that are known to be optimal in the worst case but not in general). This project is under submission to the ITCS 2023 conference.

Sep 2022 – present ***Envy-Freeness up to a Random good & Average Envy-Freeness up to Any good***, *In progress*, Advisor: Dr. Alex Psomas.

We attempt to address fair allocation's infamous problem of "Envy-Freeness up to Any good": given m indivisible items and n agents as well as their valuations for each item, we want to find a full allocation of these items to the agents such that any agent i envies any other agent j up to at most any item in j's allocated goods. In our progress to tackle this problem, we aim to solve two weaker envy-freeness criteria based on randomization, namely "Envy-Freeness up to a Random good" and "Average Envy-Freeness up to Any good".

Sep 2022 – present ***Tight Identity Testing***, *In progress*, Advisor: Dr. Paul Valiant.

Identity testing is another fundamental statistical algorithm: given samples from an unknown distribution p over n elements, an explicitly known distribution q, and parameters ϵ and δ, we wish to know if p is either exactly q or at least ϵ-far away from p in total variation distance. A common approach to this problem is to use the chi-squared hypothesis test with the null hypothesis being $p = q$ and the alternative hypothesis of $|p - q| \geq \epsilon$. We are developing an algorithm that guarantees optimal performance in all cases to replace this traditional method.

Scholarships & Awards

2022 Nominated by the Department of Computer Science for the **CRA Outstanding Undergraduate Researcher Awards**.

2021 **First Prize Winner** of the NIST "A Better Meter Stick for Differential Privacy" competition.

2020 **Bronze Medal** at ICPC North American Championship 2020.

2020 – 2022 **Qualified** for three consecutive ICPC World Finals.

2021 – 2022 Receiver of the **Jason Yu Memorial Scholarship**.

Computer skills

Programming Languages: Python, C/C++, JavaScript/TypeScript, LaTeX, Java, Haskell

Web Technologies: React/React Native

Position of Responsibility

2020 – 2022 **Head of Development of Competitive Programmers Union**, Purdue University.

2021 – 2022 **Treasurer of Competitive Programmers Union**, Purdue University.

Teaching Assistantship

Fall 2019 – present **CS390-CP2: Competitive Programming II**, Purdue University.

Spring 2020 – present **CS490-CP3: Competitive Programming III**, Purdue University.

Fall 2022 **CS381: Introduction to the Analysis of Algorithms**, Purdue University.